

RegOS Custom Work Map

Everything in this repository that is NOT stock Open WebUI. Organized by functional area.

1. Filter Functions (Open WebUI Filters)

These run inside Open WebUI as inlet/outlet filters on every chat message.

File	Version	Purpose
regos_setup/functions/graphrag_filter.py	v0.17.3	Core retrieval engine. 4-step pipeline: document fulltext, entity matching, concept expansion, direct search. Includes confidence scoring (6 signals), guardrails (keyword, zero-retrieval, jurisdiction), escalation (auto-flag + n8n webhook), threshold evaluation (regex detection of 35+ parameters against 96 limits), Sources Panel injection, conditional disclaimers. FEA v2 schema (Ch24Class/Ch24Document/Ch24Entity).
regos_setup/functions/audit_logger.py	v0.4.0	Logs every chat to SQLite (audit.db). Captures: user, chat, query, response, model, confidence score/band/signals, escalation metadata, guardrail metadata. Reads data from message dict keys set by graphrag_filter.
regos_setup/functions/escalation_action.py	v1.0.0	Manual escalation Action. User-triggered expert compliance review. Builds case packet, POSTs to n8n webhook, writes audit trail. Complement to automatic escalation in graphrag_filter.
regos_setup/functions/threshold_eval.py	---	[OBSOLETE --- unused in production] Early Tools-based threshold evaluation. Abandoned because base model (Nemotron) doesn't support tool-calling. Superseded by integrated threshold eval in graphrag_filter.py (v0.14.0). Retained for future redesign.
regos_setup/functions/graphrag_pipe.py	v0.1.0	[OBSOLETE --- abandoned] Original Pipe approach that replaced the LLM endpoint. Caused deadlock. Superseded by the Filter pattern in graphrag_filter.py (v0.3.0+).

Every .py file has a companion .md file alongside it with detailed documentation and status.

2. APIs (Sidecar Services)

Standalone FastAPI services that run alongside Open WebUI.

File	Purpose
regos_setup/api/breach_api.py	REST API for compliance dashboard --- summary, list breaches, run checks, verify evidence hashes. 7 endpoints.
regos_setup/api/scada_stream.py	SCADA streaming ingestion --- WebSocket (full-duplex), SSE (monitoring), REST batch. Rate limiting, auth, real-time threshold evaluation.
regos_setup/api/apas_bridge.py	APAS Telemetry polling bridge --- JWT auth against APAS API, metric mapping via config, unit conversion, feeds into SCADA pipeline. 30s poll interval.
regos-installer/api/regos_api.py	RegOS API wrapper used by the installer for function registration and model creation.

3. Data Files

File	Contents
regos_setup/data/regulatory_thresholds.json	96 regulatory thresholds from Chapter 24 --- concentration limits, timeframes, distances, percentages, penalties.
regos_setup/data/concepts.json	Ontology concepts for the FEA knowledge graph.
regos_setup/data/apas_metric_mappings.json	Maps APAS metric names to RegOS threshold parameters. 8 active + 14 effluent sensor templates.
regos_setup/data/chaptor_24_graph.json	Full Neo4j graph export --- 693 nodes, 2,351 relationships (FEA v2 schema).
regos_setup/cypher/batch_01.cypher	Cypher statements for batch-loading the graph into Neo4j.

Root-level copies: regulatory_thresholds.json and neo4j_query_table_data_2026-1-15 (1).json (237 MB raw graph export) also exist at root for convenience.

4. System Prompt & Prompts

File	Purpose
regos_setup/system_prompt.md	RegOS Compliance Copilot system prompt. Defines 3 personas (Citizen, Consultant, Regulator), response structure, citation rules, scope boundaries, refusal formatting, threshold evaluation instructions. Applied to the custom model.
regos_setup/prompts/graph_extraction_prompt.md	Prompt template for extracting entities/relationships from regulatory text into the knowledge graph.

5. Installer & Deployment

Directory	Purpose
regos-installer/	Current installer (v2). 10-step automated deployment. Includes source patches (apply-patches.py --- 15 patches + 3 new Svelte files), shell libraries, config, and full documentation bundle. See SOURCE_PATCHES.md for details.
gtihub-regos-installer/	Earlier installer (v1). 8 steps, no guest access or admin panel patches. Kept for reference.
regos_setup/setup/	Setup scripts and README for fresh deployment from scratch (RunPod, Docker, etc.).

6. Source Patches (Modified Stock Files)

15 patches across 13 stock Open WebUI files. Adds: RegOS admin panel, guest access with email + rate limiting, disclaimer modal, confidence display config, role-based user filtering, neo4j dependency. See SOURCE_PATCHES.md for the complete file-by-file breakdown.

7. Benchmarks

File/Dir	Purpose
benchmark/	Dockerized PDF extraction benchmark comparing Docling vs GPT-4o on the Opa-locka municipal code.
benchmark/benchmark_gpt4o.py	GPT-4o extraction script
benchmark/score_and_report.py	Scoring and report generation
benchmark/results/	Output: baseline text, Docling JSON/MD, GPT-4o per-page and combined, sample PDFs, benchmark_report.json
pdf_benchmark.py (root)	Docling benchmark runner
benchmark_strategy.md (root)	Benchmark strategy document

8. Standalone Applications

Directory	Purpose
APAS-Legal-PDF-Chunking-Dashboard/	Full-stack app for visualizing legal PDF chunking. Backend: FastAPI with chunker, enrichment, extraction, neo4j_service. Frontend: Next.js + Tailwind. Docker Compose deployment. Planned for extraction to separate repo.

9. Documentation

regos-docs/ (consolidated from root --- 33 .docx, 2 .xlsx, 1 .pptx, 1 .svg, 2 .drawio, 1 .zip, 1 .md)

All RegOS documentation files consolidated into regos-docs/. Organized by category:

Architecture & Technical: RegOS_Architecture, RegOS_API_Architecture, RegOS_API_Sample_Response, RegOS_Infrastructure_Diagram.svg, RegOS_Infrastructure_v2.drawio, RegOS_expanded.drawio, Dashboard_Technical_Overview_and_Costs

Features: RegOS_Escalation_Feature_Documentation, RegOS_Escalation_Workflow_Scope_P1_P2, RegOS_Disclaimer_GuestMode_Build_Documentation, RegOS_Neo4j_Failover_Documentation, RegOS_Security_v019_Documentation

Security: RegOS_Security_Phase1_Phase2_Implementation_Plan, RegOS_Security_Research_Backlog_Item5, RegOS_Security_Research_Revised, RegOS_Security_Test_Results

Research: RegOS_Research_Backlog_Item9_Confidence_Feedback_Loop (+ Reviewed), RegOS_PDF_Parsing_Research, RegOS_Discovery_Document

Deployment & Operations: RegOS_Setup_Guide, RegOS_Deployment_Guide, RegOS_Installer_Explanation

Knowledge Transfer: APAS_Knowledge_Transfer_Program, Module_1_The_Platform_Facilitator_Guide, RegOS_Document_Analysis_KT, RegOS_Sidecar_API_KT, RegOS_Sidecar_Testing_Guide

Demo & Presentation: RegOS_Demo_Script, RegOS_P0_Full_Demo, RegOS_SHA256_Demo, Legal_PDF_Dashboard_Demo_Script.pptx

Decision Records: Docling_Decision_Record.md (self-contained benchmark decision with embedded results)

Tracking: RegOS_P1_P2_Backlog.xlsx, RegOS_MECE_Test_Prompts.xlsx, RegOS-Changelog-2026-03-10.docx

strategies/ (strategy documents)

Municipal_Code_Chunking_Strategy.md, benchmark_strategy.md

regos_setup/extra/docs/ (technical .md docs)

confidence_scoring.md, graphrag_filter.md (extended), audit_logger.md, disclaimer_research.md, graphrag_vs_rag.md, functions_README.md, GRAPH_REBUILD_QUICKSTART.md, N8N_ESCALATION_SETUP.md, P0_Feature_Analysis.md, P0_Implementation_Order.md, demo_script.md, readme1.md

10. Utility & Demo Scripts

File	Purpose
regos_setup/scripts/	Utility scripts (setup, diagnostics)
regos-installer/scripts/demo_show_records.py	Show audit/breach DB records
regos-installer/scripts/demo_tamper.py	Tamper with evidence hash for SHA-256 demo
regos-installer/scripts/demo_reset.py	Reset demo environment
regos-installer/scripts/verify_hashes.py	Verify evidence hash integrity

Root copies of these scripts are flagged for removal (see REMOVAL_CANDIDATES.md).

11. Workflow Automation (n8n)

File	Purpose
regos_setup/n8n/regos_escalation_workflow.json	Importable n8n workflow --- webhook receives escalation case packet, AI generates case brief, stores case file + returns confirmation

File	Purpose
regos_setup/n8n/email_preview.html	HTML preview of escalation email (moved from root)
regos_setup/n8n/format_email_node_v3.js	n8n JavaScript node for email formatting (moved from root)
regos_setup/n8n/parsed_response.html	Parsed response HTML preview (moved from root)

12. Test Suites

File	Purpose
regos_setup/tests/adversarial_guardrails.md	28 test queries across 7 categories: out-of-scope, zero-retrieval, jurisdiction, citation fabrication, prompt injection, legitimate, edge cases
regos_setup/tests/phase1_phase2_test_results.md	Phase 1/2 test results
RegOS_MECE_Test_Prompts.xlsx (root)	MECE test prompt matrix

13. Deliverable Packages

Directory	Contents
RegOS_Integration_Package/	Bundled for external delivery: RegOS_Integration_Response.docx + 3 JSON data files (thresholds, concepts, mappings)
Docling_vs_GPT4o_Benchmark_Bundle.zip	Bundled benchmark results for distribution

Feature Version History

#	Feature	Version	Status
1	Audit Logging	v0.4.0	Done
2	GraphRAG Pipeline	v0.17.3	Done
3	Confidence Scoring	v0.6.0+	Done
4	Enterprise Output Formatting	v0.7.0+	Done
4b	Sources Panel Integration	v0.8.0	Done
4c	Conditional Disclaimers	v0.8.1	Done
5	Escalation Workflow + n8n	v0.10.0	Done
6	Threshold Evaluation (weights + service)	v0.13.0	Done

#	Feature	Version	Status
7b	SCADA Streaming API	v0.1.0	Done
7c	APAS Telemetry Bridge	v0.1.0	Done
8	Refusal & Guardrails	v0.12.0	Done
9	Integrated Threshold Eval	v0.14.0	Done

Full development narrative: regos_setup/REGOS_CHANGELOG.md

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